

Assignment 3 - Task 1

Multi-class Classification using Fully Connected Neural Networks

Summary

This assignment implements fully connected neural networks from scratch for three-class classification on a linearly separable Gaussian dataset and a nonlinear spiral dataset.

Dataset	Selected architecture	Validation	Test
Gaussian clusters	2-2-3	100.00%	100.00%
Spiral	2-16-16-3	99.67%	99.33%

Methods: sigmoid activation, squared error loss, manual backpropagation, stochastic gradient descent, and validation-based model selection.

1. Overview and Objectives

Two two-dimensional, three-class datasets were generated. Dataset 1 consists of compact Gaussian-like clusters and is linearly separable. Dataset 2 consists of three interleaving spiral arms and is nonlinear.

- Generate the two required three-class synthetic datasets.
- Use a 60% training, 20% validation and 20% test split for each class.
- Implement FCNNs, squared-error loss, backpropagation and SGD from scratch.
- Compare different hidden-node configurations and select the best using validation accuracy.
- Evaluate training error, decision regions, confusion matrices and node outputs.
- Compare FCNN performance with a single-neuron baseline.

2. Dataset Visualization

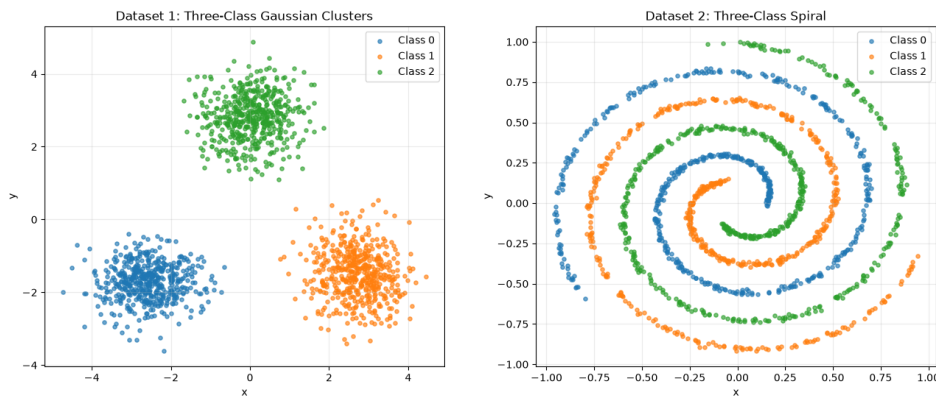


Figure 1. Gaussian-cluster dataset (left) and spiral dataset (right).

Each dataset contains 1500 samples: 500 per class. The split gives 900 training, 300 validation and 300 test samples, with 100 samples per class in validation and test sets.

3. Methodology

The input has two variables and the output layer has three nodes. Sigmoid activation is used throughout the network. The predicted class is the output node with the largest activation.

Dataset	Network structure	Architectures tested
Gaussian	$2 \rightarrow H \rightarrow 3$	$H = 2, 4, 8, 16, 32$
Spiral	$2 \rightarrow H1 \rightarrow H2 \rightarrow 3$	$(4,4), (8,8), (16,16)$

Weights are updated after every training example using stochastic gradient descent. Squared error is the instantaneous loss. Gaussian models use 1200 epochs and spiral models use 1800 epochs.

4. Dataset 1 - Gaussian Clusters

All five tested architectures achieved 100% validation accuracy. The first architecture in the tested order, 2-2-3, was selected.

Architecture	Hidden nodes	Training	Validation	Final error
2-2-3	2	100.00%	100.00%	0.000279
2-4-3	4	100.00%	100.00%	0.000092
2-8-3	8	100.00%	100.00%	0.000094
2-16-3	16	100.00%	100.00%	0.000087
2-32-3	32	100.00%	100.00%	0.000106

Best model: 2-2-3 | Validation: 100.00% | Test: 100.00%

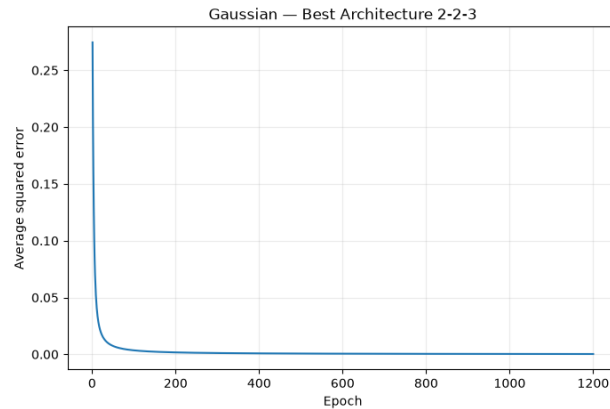


Figure 2. Average squared error versus epochs for the best Gaussian architecture.

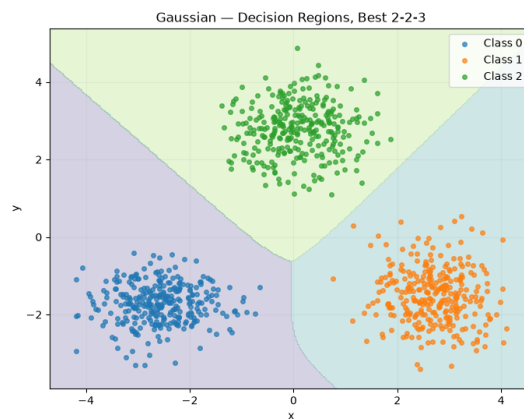


Figure 3. Gaussian decision regions with training data.

The Gaussian classes are well separated, so even the smallest tested network is sufficient for perfect classification.

5. Dataset 2 · Spiral

The selected architecture is 2-16-16-3, with 99.67% validation accuracy and 99.33% test accuracy.

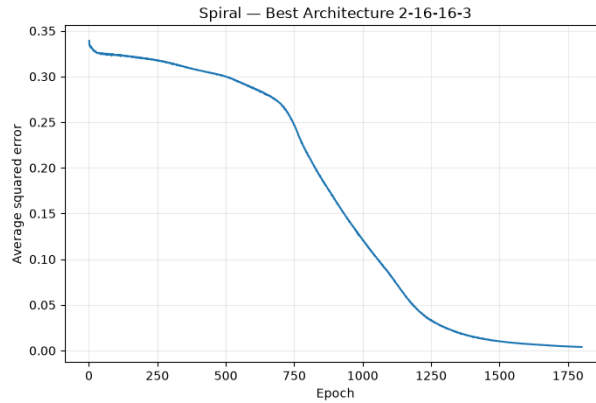


Figure 4. Average squared error versus epochs for the best spiral architecture.

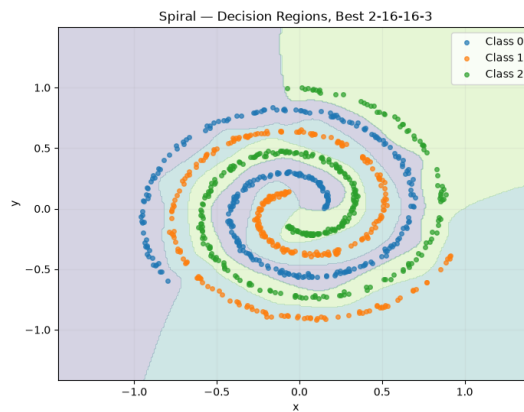


Figure 5. Spiral decision regions with training data.

6. Classification Results

Architecture	Validation	Confusion matrix
2-2-3 (Gaussian)	100.00%	100 0 0 / 0 100 0 / 0 0 100
2-4-3 (Gaussian)	100.00%	100 0 0 / 0 100 0 / 0 0 100
2-8-3 (Gaussian)	100.00%	100 0 0 / 0 100 0 / 0 0 100
2-16-3 (Gaussian)	100.00%	100 0 0 / 0 100 0 / 0 0 100
2-32-3 (Gaussian)	100.00%	100 0 0 / 0 100 0 / 0 0 100
2-4-4-3 (Spiral)	72.00%	58 18 24 / 7 87 6 / 7 22 71
2-8-8-3 (Spiral)	96.00%	98 2 0 / 2 93 5 / 2 1 97
2-16-16-3 (Spiral)	99.67%	99 0 1 / 0 100 0 / 0 0 100

Best spiral test confusion matrix

	Predicted 0	Predicted 1	Predicted 2
True 0	99	0	1
True 1	0	100	0
True 2	1	0	99

7. Single-Neuron Comparison

Dataset	Model	Training	Validation	Test
Gaussian	Single linear neuron	100.00%	100.00%	100.00%
Gaussian	Best FCNN [2]	100.00%	100.00%	100.00%
Spiral	Single linear neuron	39.78%	44.00%	38.33%
Spiral	Best FCNN [16,16]	100.00%	99.67%	99.33%

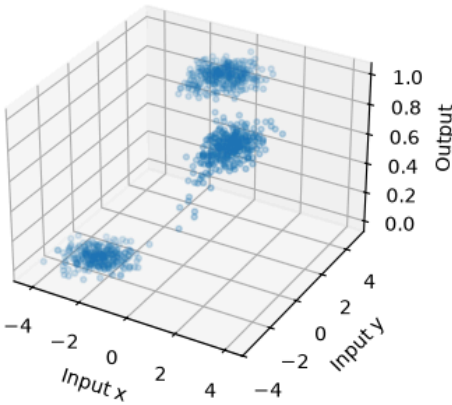
The Gaussian data are linearly separable, so the single linear neuron already achieves perfect classification. The spiral data require nonlinear decision boundaries; the single neuron performs poorly, while the best FCNN reaches 99.67% validation and 99.33% test accuracy.

10.1 Gaussian · Training: 3-D Node-Output Plots

Best architecture: 2-2-3. Hidden and output node responses are shown for the selected model.

Gaussian | Training | Hidden layer 1

Hidden layer 1 | Node 1



Hidden layer 1 | Node 2

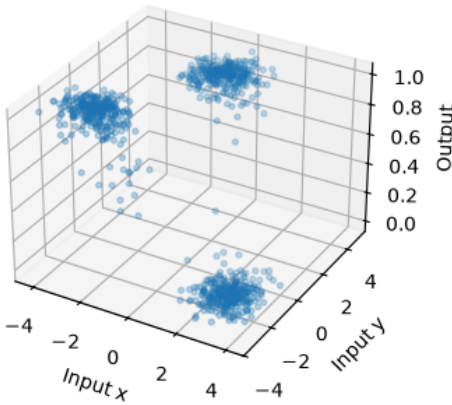
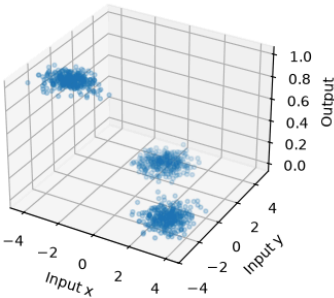


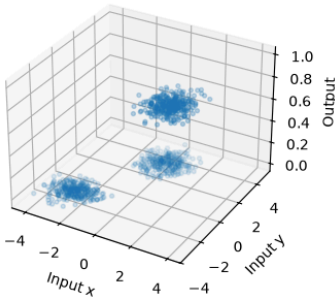
Figure. Training data · hidden-layer node outputs.

Gaussian | Training | Output

Output | Node 1



Output | Node 2



Output | Node 3

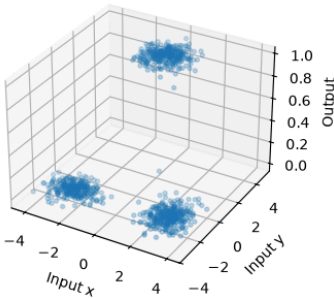


Figure. Training data · output-node outputs.

10.2 Gaussian · Validation: 3-D Node-Output Plots

Best architecture: 2-2-3. Hidden and output node responses are shown for the selected model.

Gaussian | Validation | Hidden layer 1



Figure. Validation data · hidden-layer node outputs.

Gaussian | Validation | Output

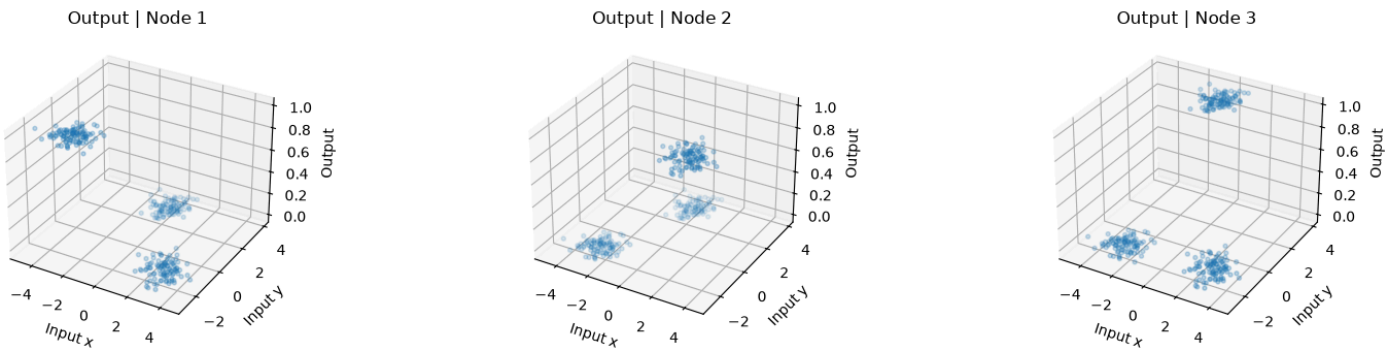


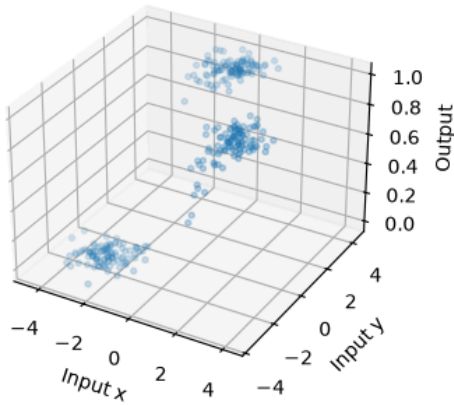
Figure. Validation data · output-node outputs.

10.3 Gaussian · Test: 3-D Node-Output Plots

Best architecture: 2-2-3. Hidden and output node responses are shown for the selected model.

Gaussian | Test | Hidden layer 1

Hidden layer 1 | Node 1



Hidden layer 1 | Node 2

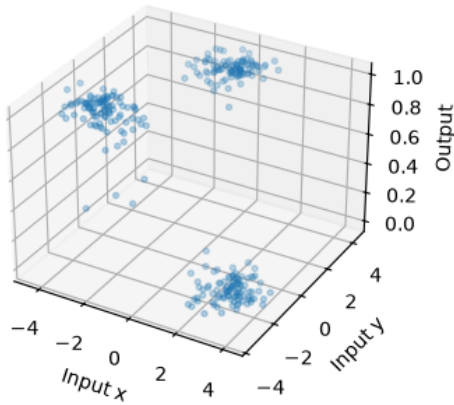
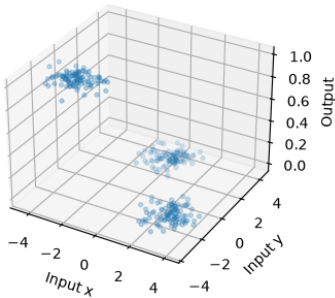


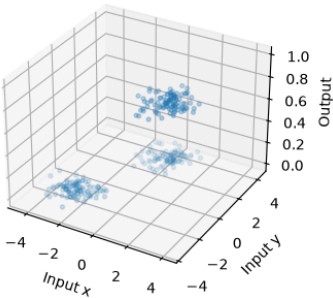
Figure. Test data · hidden-layer node outputs.

Gaussian | Test | Output

Output | Node 1



Output | Node 2



Output | Node 3

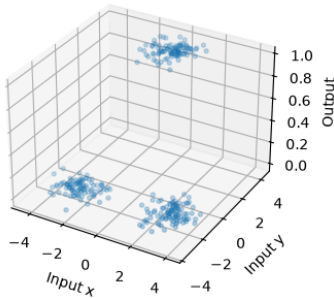


Figure. Test data · output-node outputs.

11.1 Spiral · Training: 3-D Node-Output Plots

Best architecture: 2-16-16-3. All hidden and output nodes are shown.

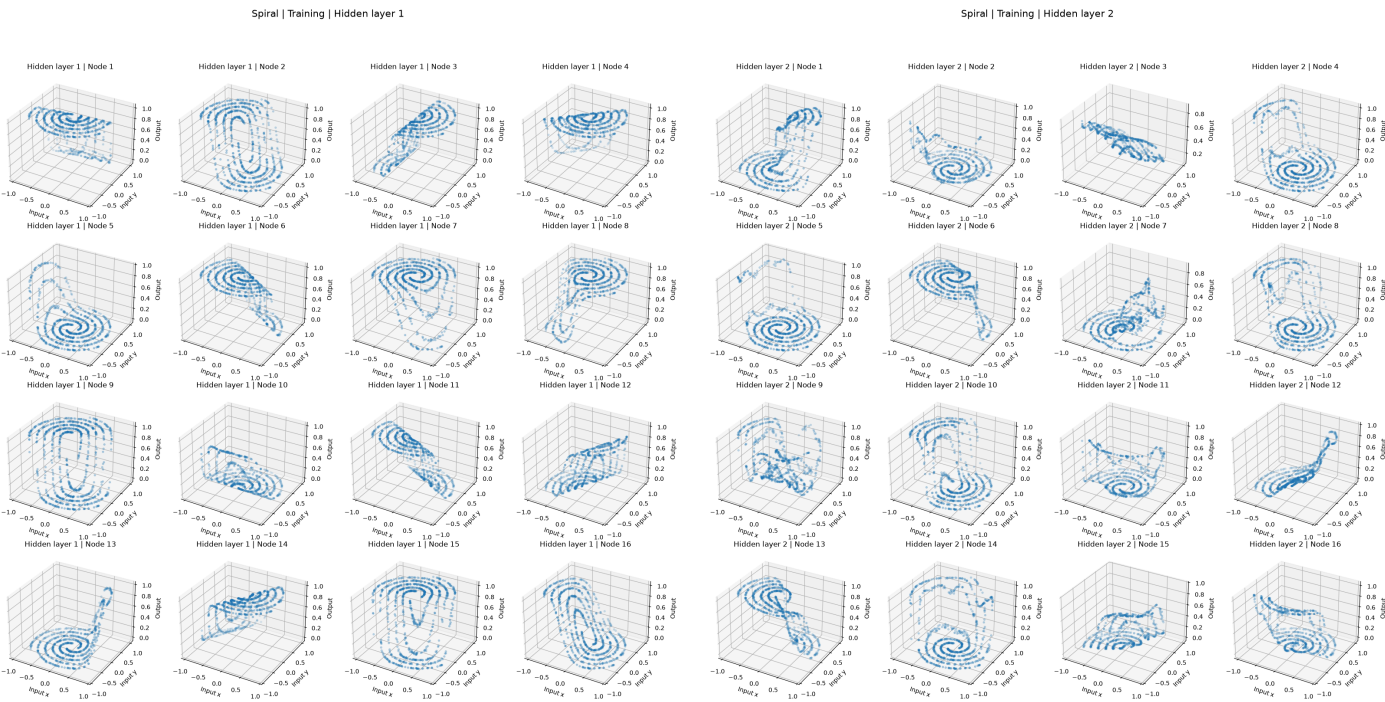


Figure. Training data · hidden layer 1 and hidden layer 2.

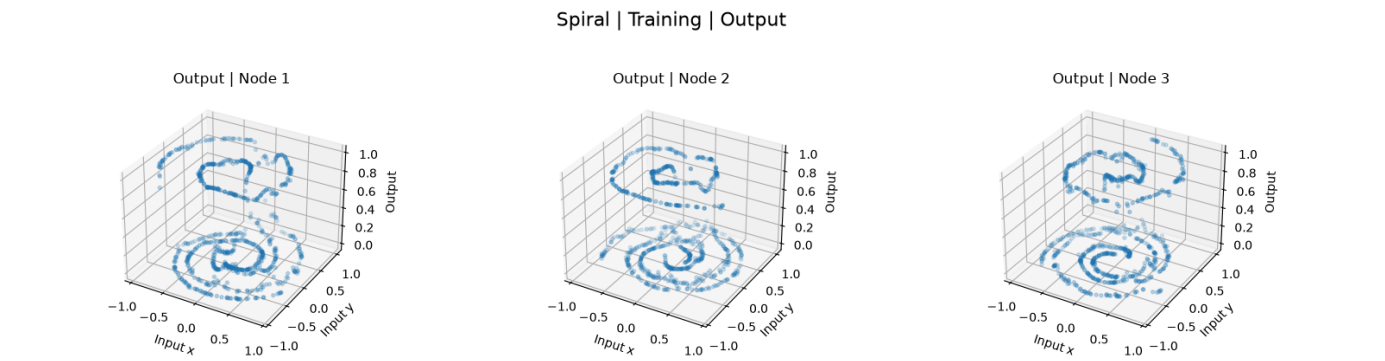


Figure. Training data · output-node outputs.

11.2 Spiral · Validation: 3-D Node-Output Plots

Best architecture: 2-16-16-3. All hidden and output nodes are shown.

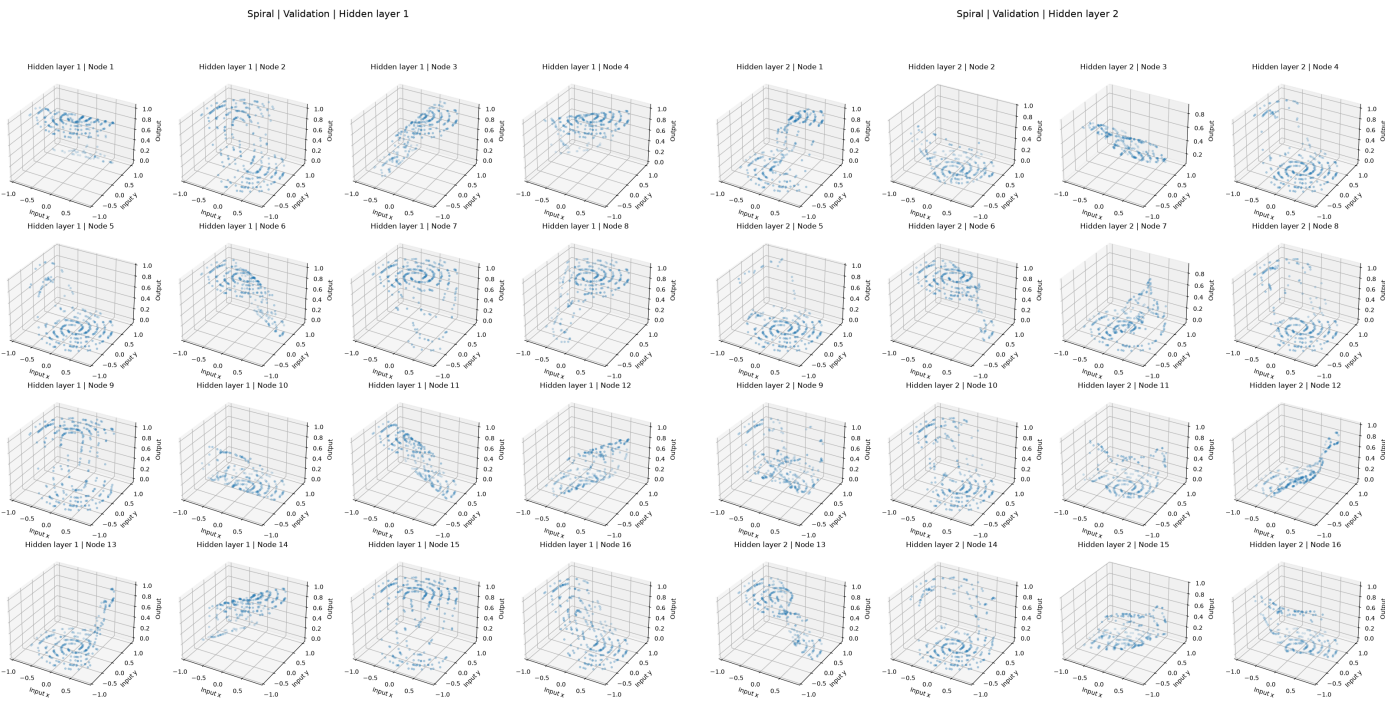


Figure. Validation data · hidden layer 1 and hidden layer 2.

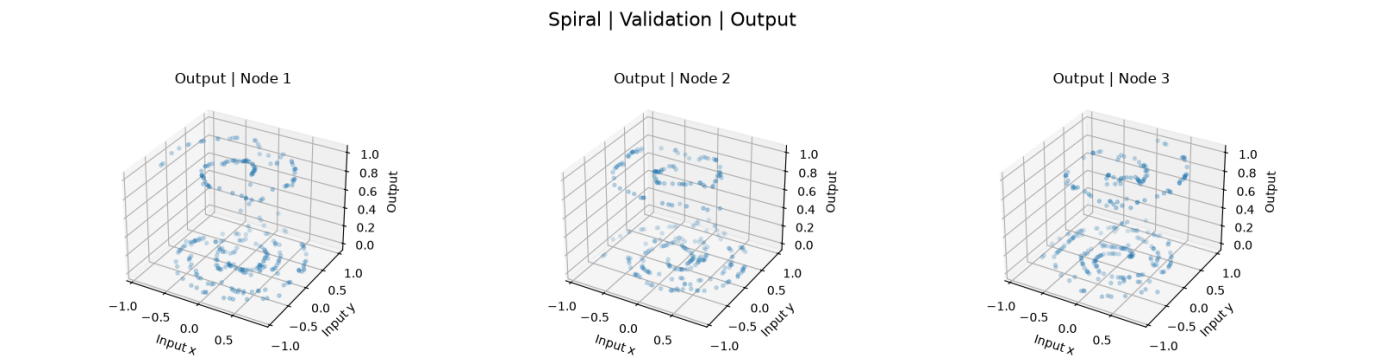


Figure. Validation data · output-node outputs.

11.3 Spiral · Test: 3-D Node-Output Plots

Best architecture: 2-16-16-3. All hidden and output nodes are shown.

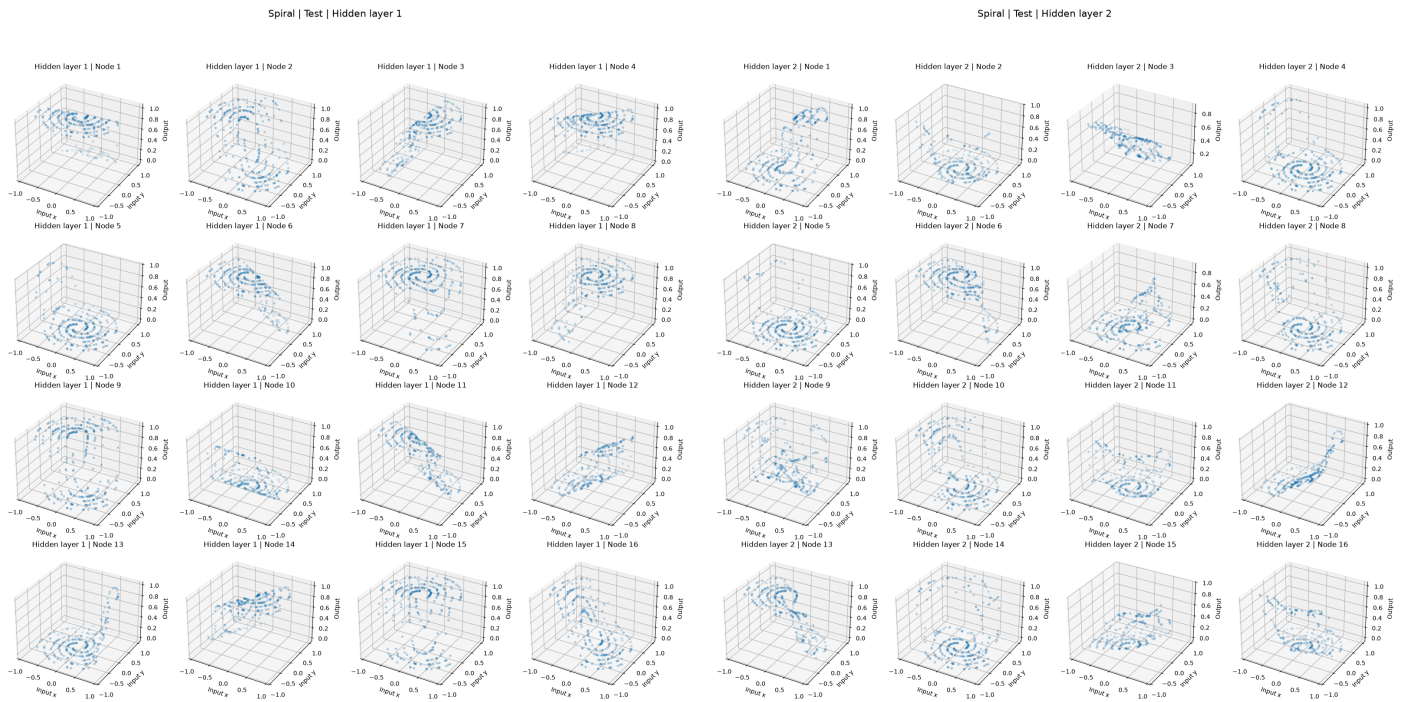


Figure. Test data · hidden layer 1 and hidden layer 2.

Spiral | Test | Output

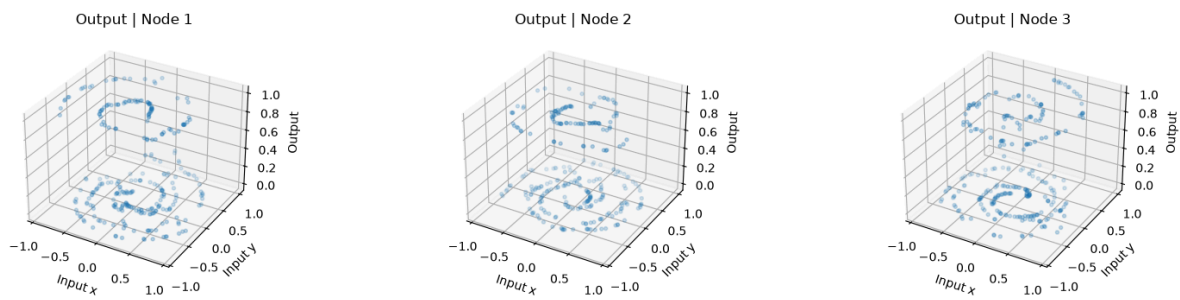


Figure. Test data · output-node outputs.

8. Inferences

- The Gaussian problem is linearly separable and needs only a small network.
- The spiral problem needs nonlinear representation and benefits from larger hidden layers.
- The selected models generalize well from validation to test data.
- The error curves show progressive learning during training.

9. Conclusion

The FCNN, backpropagation and SGD were implemented from scratch. The experiments demonstrate how network capacity affects nonlinear classification, with 2-2-3 sufficient for the Gaussian data and 2-16-16-3 providing excellent performance on the spiral data.